



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE(Advanced)
UNIT TEST
30-01-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : I(A) & LIVE PHASE-I(A)]

Time : 3 Hours

PAPER-1

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **32** pages and that all the **19** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- **Boltzmann constant** $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$
- **Coulomb's law constant** $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$
- **Universal gravitational constant** $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$
- **Speed of light in vacuum** $c = 3 \times 10^8 \text{ ms}^{-1}$
- **Stefan-Boltzmann constant** $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$
- **Wien's displacement law constant** $b = 2.89 \times 10^{-3} \text{ m-K}$
- **Permeability of vacuum** $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$
- **Permittivity of vacuum** $\epsilon_0 = \frac{1}{\mu_0 c^2}$
- **Planck constant** $h = 6.63 \times 10^{-34} \text{ J-s}$

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

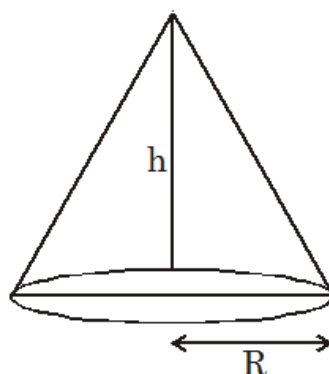
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

-
1. The maximum force on a body of mass m executing SHM is p and the maximum kinetic energy of the body is q . What is its time period?
 (A) $\frac{2\pi}{p} \sqrt{2mq}$ (B) $\frac{2\pi}{q} \sqrt{2mp}$ (C) $\frac{\pi}{p} \sqrt{2mq}$ (D) $\frac{\pi}{q} \sqrt{2mp}$
 2. Potential energy of a particle executing SHM varies sinusoidally. If the frequency of oscillation of the particle is ' f ' then frequency of oscillation of potential energy is
 (A) f (B) $\frac{f}{2}$ (C) $4f$ (D) $2f$
-

Space for Rough Work

3. A liquid of density ρ is filled in a conical vessel as shown in figure. Force exerted by liquid on side wall is :



- (A) $\frac{2}{3}\pi R^2 \rho gh$ (B) $\frac{1}{3}\pi R^2 \rho gh$ (C) $\pi R^2 \rho gh$ (D) 0
4. A small ball of density $4\rho_0$ is released from rest just below the surface of a liquid. The density of liquid increases with depth as $\rho = \rho_0 (1 + ay)$ where $a = 2\text{m}^{-1}$ is a constant. Find the time period of its oscillation. (neglect the viscosity effects)

- (A) $\frac{2\pi}{\sqrt{5}}\text{sec}$ (B) $\frac{\pi}{\sqrt{5}}\text{sec}$ (C) $\frac{\pi}{2\sqrt{5}}\text{sec}$ (D) $\frac{3\pi}{2\sqrt{5}}\text{sec}$

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. In a simple harmonic motion of amplitude A (symbols have their usual meaning)

(A) Time taken by particle to move from mean position to $\frac{A}{2}$ is equal to time taken by particle to move from $\frac{A}{2}$ to A

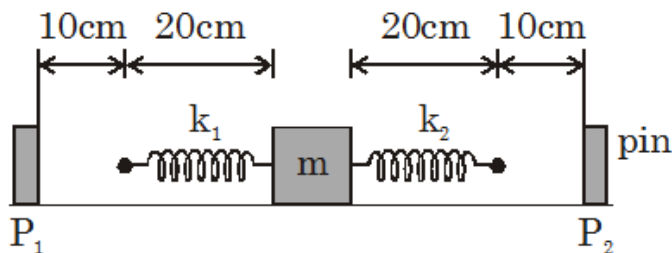
(B) Average velocity of particle in one time period T is $\frac{4A}{T}$

(C) Net work done on block in one time period T is 0

(D) Particle has maximum kinetic energy at mean position

Space for Rough Work

6. Two spring, each of outstretched length 20 cm but having different spring constants $k_1 = 1000\text{N/m}$ and $k_2 = 3000\text{ N/m}$ are attached to two opposite faces of a block of mass $m = 100\text{ g}$ kept on a smooth horizontal surface as shown in figure.



The outer ends of the two springs are now attached to two pins P_1 and P_2 whose locations are shown in the figure, as a result of this, the block acquired a new equilibrium position. The block has been displaced by small amount from its equilibrium position and released to perform simple harmonic motion, then.

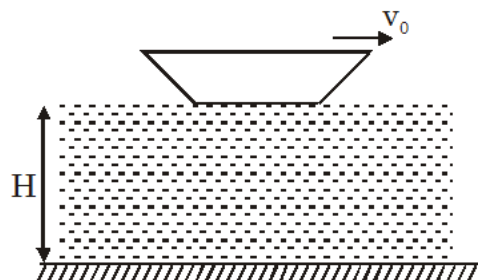
- (A) New equilibrium position is at 35 cm from P_1 and time period of simple harmonic motion is $\frac{\pi}{100}\text{ s}$.
- (B) New equilibrium position is at 20 cm from P_1 and time period of simple harmonic motion is $\frac{\pi}{100}\text{ s}$.
- (C) New equilibrium position is at 35 cm from P_1 and time period of simple harmonic motion is $\frac{\pi}{26}\text{ s}$.
- (D) New equilibrium position is at 30 cm from P_1 and time period of simple harmonic is $\frac{\pi}{26}\text{ s}$

Space for Rough Work

7. A particle of mass m moves along X-axis. If v is the velocity at a distance x from a origin on the X-axis such that $v^2 = a - bx^2$, where a and b are constant, then (Assume that potential energy of the particle is zero when it is at origin)
- (A) the particle continues to move along the positive x-direction only
- (B) the motion is simple harmonic
- (C) the particle oscillates with a frequency equal to $\frac{\sqrt{b}}{2\pi}$
- (D) the total energy of the particle is $\frac{ma}{2}$

Space for Rough Work

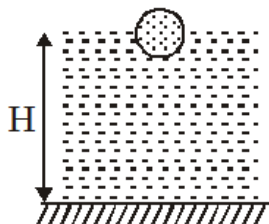
8. Water is flowing on a horizontal surface where velocity varies with height y from the ground as $v = \frac{v_0}{H^2} \left(2y^2 - \frac{y^3}{H} \right)$ where H is the depth of water. Coefficient of viscosity of water is η . Then choose the correct option(s).



- (A) Shear stress at $y = \frac{H}{2}$ is $\frac{5v_0\eta}{4H}$
- (B) Shear stress at $y = H$ is $\frac{\eta v_0}{H}$
- (C) Viscous force acting on the boat of base area A is $\frac{\eta A v_0}{H}$
- (D) Viscous force acting on the boat of base area A is $\frac{\eta A v_0}{2H}$

Space for Rough Work

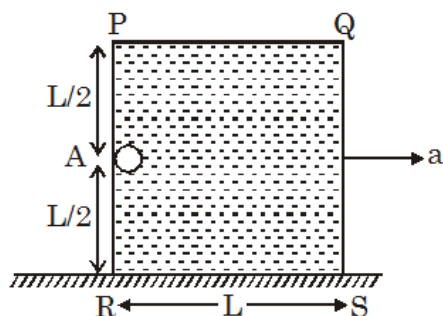
9. A sphere of radius r and density ρ is released from the surface of a liquid of density $\frac{\rho}{2}$ and coefficient of viscosity η . It achieves terminal velocity just before reaching the ground. Then choose the correct option(s).



- (A) Terminal velocity is $\frac{2r^2\rho g}{9\eta}$
- (B) Terminal velocity is $\frac{r^2\rho g}{9\eta}$
- (C) Magnitude of work done by viscous force on the sphere is $\frac{2\pi r^3\rho g}{3} \left(H - \frac{\rho^2 r^4 g}{81\eta^2} \right)$
- (D) Magnitude of work done by viscous force on the sphere is $\frac{4}{3}\pi r^3\rho g \left(H - \frac{\rho^2 r^4 g}{9\eta^2} \right)$

Space for Rough Work

10. A small solid ball of density ρ is held inside at point A in a closed cubical container of side L , filled with an ideal liquid of density 4ρ as shown in the figure. Now, if the container starts moving with constant acceleration a horizontally and the ball is released from point A simultaneously, then



- (A) For ball to hit the top of container at end Q, $a = 3g$
 (B) For ball to hit the top of container at end Q, $a = 2g$
 (C) Ball hits the top of container at end Q after a time $t = \sqrt{\frac{L}{3g}}$
 (D) Ball hits the top of container at end Q after a time $t = \sqrt{\frac{2L}{3g}}$

Space for Rough Work

SECTION-II (i) : (Maximum Marks: 12)

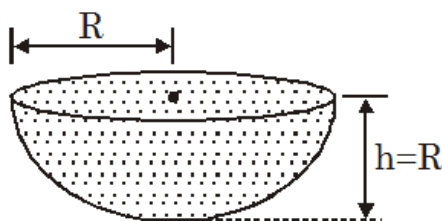
- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

A hemispherical tank of radius R is completely filled with water. Now an orifice of small area 'a' is made at the bottom of tank. (Given that $R = 10$ m, $a = 1 \text{ mm}^2$)



1. Time rate of volume emission from orifice when tank have $\frac{h}{2}$ height of water level is (in cm^3/sec)
2. The time required to empty the tank is $n \times 10^6$ sec. Here n is –

Space for Rough Work

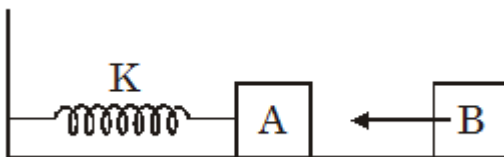
Paragraph for Questions 3 and 4

A small object of mass 2 kg is placed on the x-axis in a potential field which potential energy is define as $U = 16 (x^2 - 2x)$ where position on x-axis, x is in meter and potential energy U is in joule. If the object is released from origin i.e. $x = 0$. Then answer the following questions.

3. Angular frequency of oscillations (in rad/second) is :
4. Maximum kinetic energy (in joule) during oscillations is :

Paragraph for Questions 5 and 6

A block A of mass 1 kg connected with a spring is executing SHM. The displacement time equation of block is $x = 8 + 4 \sin (10t)$ (here x is in m). An identical block B moving towards negative x-axis with velocity 10 m/s collides with block A and sticks to it at time $t = 0$



5. Speed of block A (in m/s) just after the collision is
6. New amplitude (in m) of oscillation of block A is

Space for Rough Work

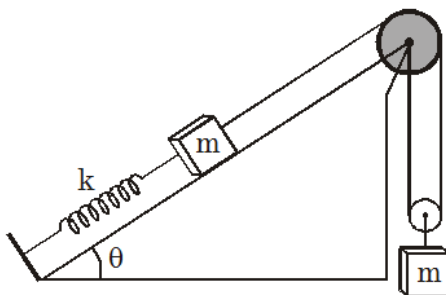
SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

7. A spring of force constant $k = 300 \text{ N/m}$ connects two blocks having masses 2 kg and 3 kg , lying on a smooth horizontal plane. If the spring block system is released from a stretched position, the number of complete oscillations in 1 minute is $6 \times N$, find the value of N . Take $\pi = \sqrt{10}$.
8. A block of mass m hangs from a vertical spring of spring constant k . If the block is pulled down by a distance of 3 mg/k from its equilibrium position and released, for the subsequent SHM of the block the ratio of maximum elongation in the spring to maximum compression in it is
9. If natural frequency ω of the system shown in figure is $x\sqrt{\frac{k}{5m}}$, find x .



Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
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- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. Match the types of defect given in Column-I with the statement given in Column-II.

Column-I		Column-II	
(i)	Impurity defect	(a)	NaCl with anionic sites called F-centres
(ii)	Metal excess defect	(b)	FeO doped with Fe^{3+}
(iii)	Metal deficiency defect	(c)	NaCl doped with Sr^{2+} and some cationic sites vacant

(A) (i)-a, (ii)-b, (iii)-c

(B) (i)-c, (ii)-a, (iii)-b

(C) (i)-a, (ii)-c, (iii)-b

(D) (i)-b, (ii)-c, (iii)-a

2. Which of the following structure is non-planar and have atleast one 'Cl' as bridging atom?

(A) Be_2Cl_4

(B) Al_2Cl_6

(C) Si_2Cl_6

(D) I_2Cl_6

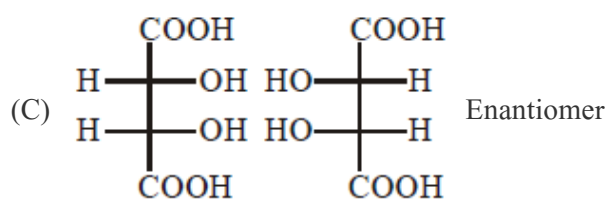
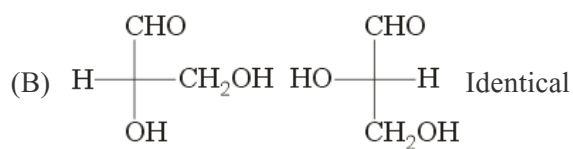
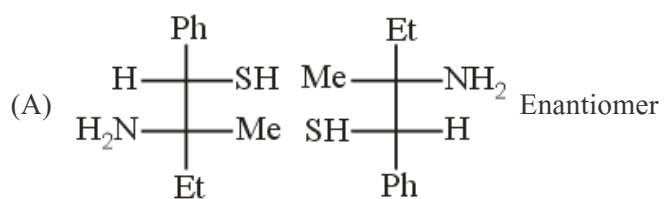
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3.



Space for Rough Work

4. Which one is correct in a given isomeric pair :



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SECTION-I (ii) : (Maximum Marks: 24)

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Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

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- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

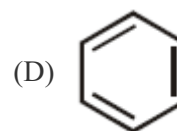
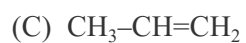
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5. A cubical crystal is such that the corners atoms are occupied by P atoms, face centers are occupied by Q atoms, alternate tetrahedral voids are occupied by R atoms and edge centers are occupied by S atoms. Identify the correct option(s) :
- (A) If there are no defects in the crystal, then the formula should be $PQ_3S_4R_3$
- (B) If all the atoms lying on one of the body diagonal are missing then formula should be $PQ_4, R_{8/3}S_4$
- (C) If all the atoms lying on a line passing through edge centers and face-centre of a face are missing, then formula should be $P_2Q_5R_8S_5$
- (D) If all the atoms lying on a line passing through body center and connecting opposite face centers are missing, then formula will be $PQ_2R_4S_3$
-

Space for Rough Work

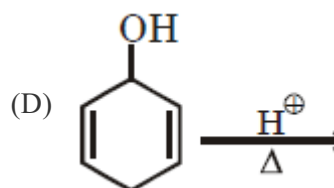
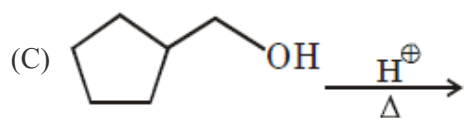
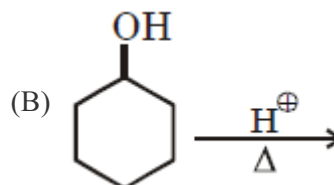
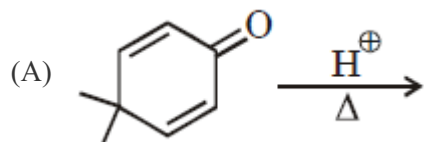
6. Which of the following statements are correct -
- (A) Amorphous solids are also called pseudo solids or supercooled liquids.
- (B) If a solid is very hard, and it is electrical insulator in solid as well as in molten state and melts at extremely high temperature, then it can be covalent or network solid
- (C) In Rhombohedral crystal system, only primitive unit cell is possible
- (D) In fluorite structure coordination number of cation is 4.
7. Consider four configurations of B_2 and select correct statements(s)-
- (I) $\sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \sigma 2p_x^2$
- (II) $\sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \pi 2p_y^2$
- (III) $\sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \pi 2p_x^1 = \pi 2p_y^1$
- (IV) $\sigma 1s^3, \sigma^* 1s^3, \sigma 2s^3, \sigma^* 2s^1$
- (A) (I) is ground state configuration
- (B) (II) violate Hund's rule
- (C) (III) violate s-p mixing
- (D) (IV) violate Pauli exclusion principle

Space for Rough Work

8. Which of the following is/are nucleophile

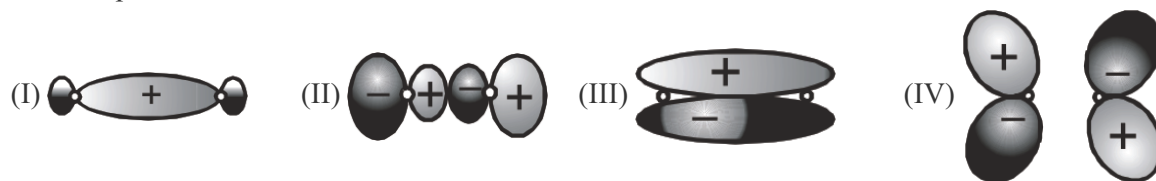


9. Which of the following reaction(s) gives aromatic product as major product :



Space for Rough Work

10. Consider following molecular orbitals (MO) which are formed by LCAO of 2p-2p orbitals and select correct option



- (A) MO-IV is a antibonding molecular orbital
 (B) In F_2 MO-IV is the HOMO & MO-II is the LUMO
 (C) In case of O_2 one unpaired electron in MO-III and other one in MO-IV respectively
 (D) Energy of molecular orbitals in O_2^{2-} is $I < III < IV < II$

Space for Rough Work

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

A hypothetical metal M (Atomic weight = 450) crystallizes as a cubic crystal such that metal atoms touch each other along body diagonal of cube. Edge length of cubic unit cell is 1nm.

If this metal is melted then its density in molten form is 1.2 gm/cc. ($N_A = 6 \times 10^{23}$)

1. Density of solid metal in gm/cc is -
2. Percentage of empty space in molten state of metal will be - [Use : $\pi\sqrt{3} = 5.44$]

Paragraph for Questions 3 and 4

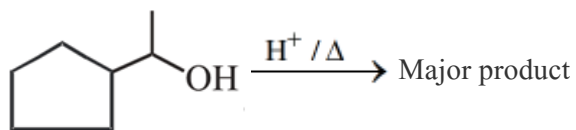
In the heavier elements of p-block elements, however as we expect from the "inert pair effect, the lower oxidation state becomes more common on descending the group." Covalent nature in ionic compound is explained on the basis of polarisation effect.

3. Atomic number of metal in the compound having most covalent character among the $ZnCl_2$, KCl , $CaCl_2$ and $NaCl$?
4. Among the Tl^{+x} , Pb^{+x} , Bi^{+x} , Sn^{+x} the maximum possible value of x will be.
(x is most stable oxidation state of that element)

Space for Rough Work

Paragraph for Questions 5 and 6

Dehydration of alcohol is an elimination reaction, which may proceed via formation of carbocation intermediate & carbocation rearrangement is also considered if required. Alkene is the product of the reaction where usually more stable alkene is major product.



5. Number of α -H present in major product :
6. Number of T.S. (transition state) in given reaction for major product :

Space for Rough Work

SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

7. A crystalline solid of a pure substance has a face-centred cubic structure with a cell edge of 200 pm. If the density of the substance in the crystal is 8 g cm^{-3} , then the number of atoms present in 32g of the crystal is $N \times 10^{24}$. The value of N is
8. If radius of M^+ is 113 pm and radius of X^- is 135 pm then the expected value of coordination number of X^- in $MX(s)$ is ?
9. Which of the following pair having correct order of covalent character
 $(\text{AlCl}_3 > \text{MgCl}_2)$, $(\text{BeCl}_2 > \text{LiCl})$, $(\text{PbCl}_4 > \text{PbCl}_2)$, $(\text{HgS} > \text{ZnS})$, $(\text{NiCl}_2 > \text{CaCl}_2)$

Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. Domain of function $f(x) = \log_2 \sin(x) + \sqrt{4 - x^2}$ is -
 (A) $(0, 2]$ (B) $(1, 3]$ (C) $(1, \pi]$ (D) $[-2, 2]$
2. Value of $\sin^{-1} \sin(7) + \cos^{-1} \cos(5)$ is -
 (A) $12 - 4\pi$ (B) $4\pi - 12$ (C) -2 (D) 2

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3. Let $f : [-1, 3] \rightarrow [-8, 72]$ be defined as $f(x) = 4x^3 - 12x$ then f is -
- (A) Injective & surjective
(B) Not injective but surjective
(C) Injective but not surjective
(D) Neither injective nor surjective
4. The complete set of value(s) of 'p' so that both the roots of the equation $(p - 5)x^2 - 2px + (p - 4) = 0$ are positive, one is less than 2 and other is lying between 2 & 3, is
- (A) $\left(\frac{49}{4}, 24\right)$
(B) $(5, \infty)$
(C) $(-\infty, 4) \cup \left(\frac{49}{4}, \infty\right)$
(D) None of these

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SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. If $f(x) = \sqrt{1-x^2} (\cos^2(\tan^{-1}x) - x^2 \sin^2(\tan^{-1}x))$, then -

(A) Range of $f(x)$ is $\mathbb{R}$

(B) Range of $f(x)$ is $[0, 1]$

(C) $f\left(\frac{\sqrt{3}}{2}\right) = \frac{1}{8}$

(D) $f\left(\frac{\sqrt{3}}{2}\right) = \frac{1}{4}$

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6. If a, b, c are distinct numbers in A.P., then both the roots of the quadratic equation $(a + 2b - 3c)x^2 + (b + 2c - 3a)x + (c + 2a - 3b) = 0$ are
- (A) real (B) positive
(C) negative (D) rational
7. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ and $f(x) = (x - 1)(x - 2)(x - 3) \dots (x - 101)$, then -
- (A) $f(x)$ is many one function
(B) $f(x)$ is onto function
(C) $f(x)$ is into function
(D) $f(x)$ is bounded function
8. Let $\alpha = 2\tan^{-1}\left(\frac{1}{2}\right) + \sin^{-1}\frac{3}{5}$ and $\beta = \sin^{-1}\left(\frac{12}{13}\right) + \cos^{-1}\left(\frac{4}{5}\right) + \cot^{-1}\left(\frac{16}{63}\right)$ be such that $2\sin\alpha$ and $\cos\beta$ are roots of the equation $x^2 - px + q = 0$, then
- (A) $p - q = 3$ (B) $p + q = 3$
(C) $p^2 - 2q = 5$ (D) $3p + 4q = 0$

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9. If the function $f(x)$ and $g(x)$ defined on $\mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x) = \begin{cases} -x + 5 & ; \quad x \in \text{Rational} \\ 3x & ; \quad x \in \text{Irrational} \end{cases}$$

$$g(x) = \begin{cases} x + 3 & ; \quad x \in \text{Rational} \\ 3x + 1 & ; \quad x \in \text{Irrational} \end{cases}$$

Then $h : \mathbb{R} \rightarrow \mathbb{R}$, $h(x) = f(x) + g(x)$ is -

- (A) One-one function (B) Onto function
(C) Into function (D) Many one function
10. Let $f : \mathbb{I} \rightarrow \mathbb{I}$ defined as $f(x) = 2\sin^2(2\pi x) - 10 \tan^2 5\pi x + 7 \cos(4\pi x) + 3$, then which of the following statement(s) is/are true -
- (A) $f(x)$ is periodic function
(B) $f(x)$ is an even function
(C) $f(x)$ is an odd function and it's inverse exist
(D) $f(x)$ is bounded function

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SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

Consider $f : [-1, 1] \rightarrow [3, 5]$ be a linear polynomial increasing function, then

1. $f\left(-\frac{1}{2}\right) + f\left(\frac{1}{2}\right) =$
2. $f^{-1}\left(\frac{15}{4}\right) =$

Paragraph for Questions 3 and 4

It is given that $\alpha, \beta, \gamma, \delta$ are the zeroes of the expression $f(x) = x(x-2)(x-4)(x-6) + 16$, then

3. $(4-\alpha)(4-\beta)(4-\gamma)(4-\delta)$ is equal to -
4. The smallest integer in the range of $f(x)$ is -

Paragraph for Questions 5 and 6

Consider $f(x) = \sin^{-1}x + \tan^{-1}x + \sqrt{x+1}$.

5. Number of integral values of k such that $f(x) = k$ has atleast one real solution is equal to -
6. Number of solutions of the equation $f(|x|) = 2$ is equal to -

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SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

7. If a functional equation $f(x+1) + f(x-1) = 2f(x)$, where $f(0) = 0$ and $f(x)$ is not identically zero for all $x \in \text{non-negative integer}$. If $\sum_{r=1}^n f(r) = 55 f(1)$, then value of n is equal to -
8. The sum of real roots of equation $x^2 + |x+3| + |x-3| = 24$ is
9. Let $f(x) = \begin{cases} x+1 & ; -1 \leq x \leq 0 \\ 3-x & ; 0 < x \leq 1 \end{cases}$ and $g(x) = \begin{cases} 3x+1 & ; 0 \leq x \leq 1 \\ 4x & ; -1 \leq x < 0 \end{cases}$, then the number of integers in the range of $g \circ f(x)$ is(are)_____

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NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them._____
Signature of the CandidateI have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same._____
Signature of the Invigilator

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